

Development of a virtual reality system that generates prism adaptation for multiple motor units

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Abstract

Conventionally, the adaptive body area for prism adaptation (PA) has been limited to the visible area of prism glasses, especially the unilateral upper limbs. In this study, we developed a virtual reality system that extends PA's adaptive body area. This system sets the visible area of the target and introduces a symbolic representation of the body position to realize a head position adaptation task. As a result, the developed system generates PA not to individual body parts but to a collection of multiple motor units. Furthermore, experimental results in healthy subjects showed that the head position adaptation task could generate PA in the upper trunk system (The upper trunk system is defined as a system consisting of the upper trunk, head, and upper extremities) and displace it to the left anteriorly. Therefore, the head position adaptation task is expected to be effective for the postural correction of USN patients with upper trunk system tilt to the right posteriorly. Moreover, the developed system has potential applications in a wide range of fields, such as detecting cerebellar disorders.

KEYWORDS

prism adaptation, reaching, rehabilitation, unilateral spatial neglect, virtual reality

1 | INTRODUCTION

Unilateral spatial neglect (USN) is known as a higher-order brain dysfunction that occurs after a stroke. USN is a symptom in which attention towards the side of space opposite to brain damage decreases and perception of objects becomes dull or impossible (Figure 1).^{1–3} Damages to the right brain hemisphere cause usually heavier USN symptoms and have worse prognoses as compared to the left hemisphere.^{2,3} Patients with USN symptoms experience serious difficulties in activities of daily living (ADL) by not noticing objects on the neglect side; they may fail to eat food on the neglect side of a plate, get lost by missing doors, or are injured by obstacles. Therefore, many of the everyday issues experienced by USN symptoms are serious, whereas they have neither insight into their symptoms nor awareness of the need for rehabilita-

tion. In addition, there are many reports about postural problems in USN patients^{4–6}; specifically, the head looks toward the non-neglect side, while the head and the trunk tilt back on the non-neglect side (Figure 2). Such a postural tilt and rotation toward the non-neglect side is a factor in the steering patient's vision toward the non-neglect side (Figure 1). When someone manually corrects a USN patient's posture, the trend to tilt and rotation to the non-neglect side becomes even stronger. Postural imbalance is also often seen in hemiplegia patients without USN symptoms, but normal posture and balance can be recovered by performing repetitive reaching movements of the upper trunk system in the opposite direction in which the trunk tilts.⁷ However, in the case of USN patients, the opposite direction of trunk tilt is the neglect side; it is impossible to perform a reaching task to the neglect space for USN patients.

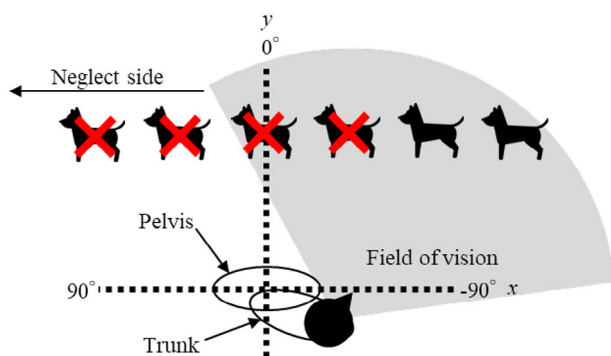


FIGURE 1 Schematic diagram of spatial cognition and posture of patients with left USN. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]



FIGURE 2 Sitting posture of a patient with USN (this photo shows a healthy person simulating a patient with left USN⁵). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Multiple approaches, including visual and auditory guidance, were proposed as rehabilitation techniques to improve spatial cognition in USN patients. However, most attempts of this kind result in improvement of specific tasks, while no significant improvement is observed for other tasks.^{1,5} The significance of rehabilitation training is very limited unless its results extend to ADL. Due to this weakness of the generalization effect, USN rehabilitation was considered difficult. In recent years, Rossetti et al. reported that prism adaptation (PA) tasks performed for about 5 min. had a generalization effect on figure-copying tasks⁸ (PA therapy will be explained in detail in Section 2). In that study, USN patients' spatial cognition improves immediately after the end of PA (POST in Figure 3), in addition to continuously improving until 2 hours later without training (LATE in Figure 3). These rehabilitation results were unprecedented in the generalization effect for non-trained tasks and the lasting improvement of spatial cognition. Since the report on USN rehabilitation was published by Rossetti et al., PA has drawn attention in the context of rehabilitation to trigger many further stud-

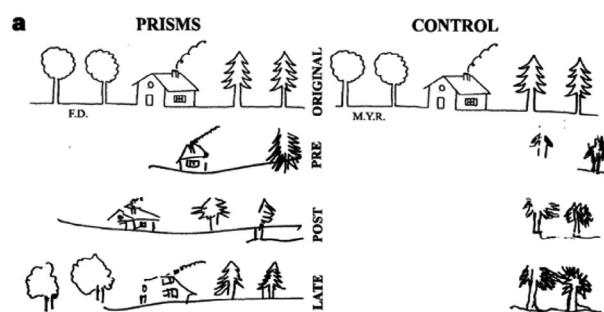


FIGURE 3 Alleviation of left USN with PA.⁸

ies. Mizuno et al. found that USN rehabilitation using PA had a generalization effect on ADL.⁹ These results showed that PA is a very promising technique for USN rehabilitation, and many further studies were conducted on the synergistic effects of PA combined with conventional USN rehabilitation methods¹ and on the clarification of the PA mechanism in experiments with healthy subjects.^{10–12} However, the body parts to which PA applies are limited to visible limbs through prism glasses, and researchers have focused on the highly dexterous upper limbs. Thus, PA could not be applied in principle to the head and other body parts that were invisible through prism glasses.

Recently, research has been conducted on virtual prism adaptation (VPA) using virtual reality (VR) instead of prism glasses.^{11,13,14} This made it possible to implement PA by means of VR devices offering a lower price and more accurate measurement instead of expensive optical devices not provided with measurement functions. However, the applicable body parts were still limited to the upper limbs. We previously proposed the expansion of VPA to head rotation movement.¹⁵ However, the VPA proposed in our previous study did not apply to reach movements of the upper trunk system involved in the postural correction of USN patients. Therefore, in the present study, we developed a VR system to apply PA to the upper trunk system. Then, we conducted experiments with the head position adaptation using healthy subjects and analyzed the factors of the prism adaptation/de-adaptation rate. Finally, we analyzed the subjects' postures during the head position adaptation and discussed the application of the proposed system for postural rehabilitation in USN patients.

2 | PRISM ADAPTATION THERAPY

2.1 | Principle of PA

PA is an adaptation to a deflected visual field using prism glasses, first reported in 1928 by Brown et al.¹⁶ Publication of the work on the application of PA to USN rehabilitation by Rossetti et al.⁸ initiated extensive research in the

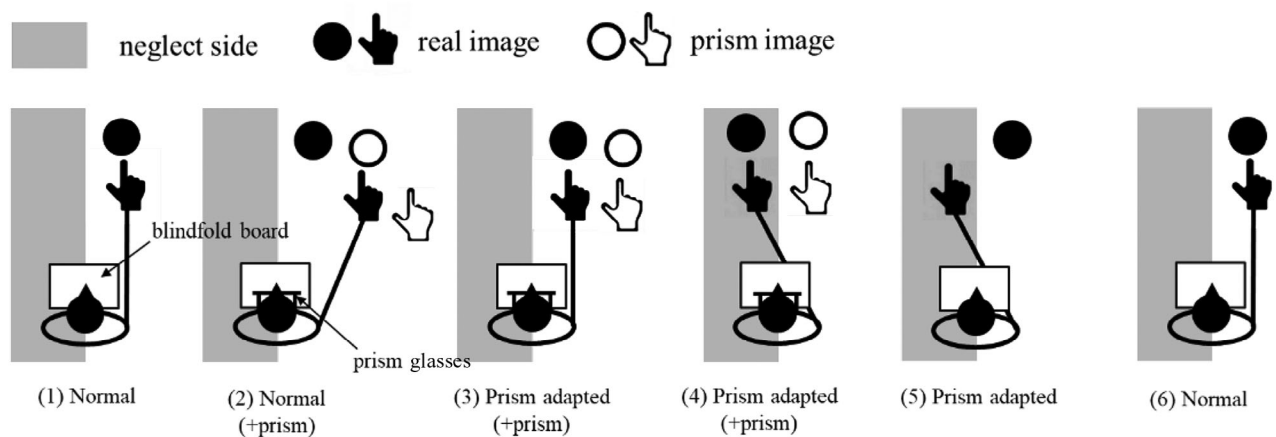


FIGURE 4 Principle of USN rehabilitation with prism adaptation (this figure is modified from Reference 15).

field of rehabilitation. This section explains the principle of PA therapy for USN rehabilitation (to modify and complement our previous paper¹⁵). Left USN is used here as an example for the explanation. A blindfold board is placed under the subject's jaw to prevent him from seeing their upper limb movement. The subject points to a target, as quickly as possible, a specified number of times without prism glasses, then with the glasses, and finally without glasses again (Figure 4 (1)–(6)). The following adaptation and de-adaptation phenomena occur during this series.

(1) Without prism (normal)

The subject points to the target correctly.

(2) With prism (normal)

The subject sees a virtual target shifted to the right (non-neglect side) through the prism and reaches his hand there. Similarly, the subject sees not his real hand but a virtual hand; thus, the subject realizes that the virtual hand is shifted to the right (non-neglect side) relative to the virtual target.

(3) With prism (adapted)

The subject gets to reach his hand to the left of the virtual target so that the virtual hand points to the virtual target precisely. This is called prism adaptation.

(4) With prism (adapted)

Due to prism adaptation, the subject reaches its hand to the left of the target (neglect side).

(5) Without prism (adapted)

The effect of the prism adaptation persists even after the glasses are removed, and the subject continues to reach his hand to the left of the target (neglect side) for a while.

(6) Without prism (normal)

The patient points to the target correctly again. This is called prism de-adaptation.

In the states of prism adaptation ((3) to (5)), the patient performs upper limb activity in the neglect space usually not performed, and this activity is thought to improve the subject's spatial cognition.

2.2 | Open-loop control in prism adaptation task

In the prism adaptation task, the subject acts as quickly as possible with the blindfold board; thus, the subject is prevented from seeing movements of his upper limbs, in this situation subject controls his upper limb in an open loop control. The subject sees a position error between his hand and target after the end of a reaching movement and adjusts his next movement. In the course of repeated trials, the subject integrates visual space deflected by the prism glasses and somatosensory space. Therefore, forcing the subject to control their motion in the open loop is a requirement for PA.

3 | VR IMPLEMENTATION OF PRISM ADAPTATION

3.1 | Definition of terms

In this study, virtual PA is realized using a virtual device instead of a prism. Consequently, the generalized

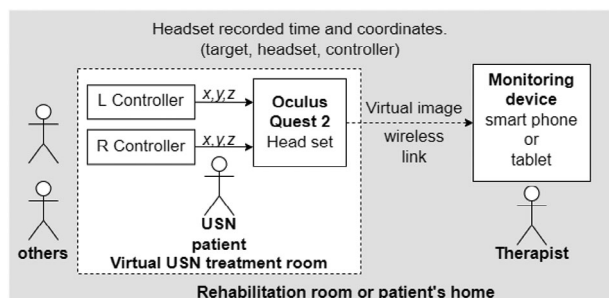


FIGURE 5 Block diagram of virtual prism adaptation system.¹⁵

terms “shift adaptation” and “shift de-adaptation” are used below rather than “prism adaptation” and “prism de-adaptation.”

3.2 | System configuration

The system configuration is shown in Figure 5. The VR headset Oculus Quest 2 was used in combination with Software Development Kit Ver. 6.01 and Unity Ver. 2021.2.2. A smartphone with Android Ver. 6.01 was used as the external monitoring device. The hardware configuration is the same as in our previous study.¹⁵ In this system, xyz coordinates of the controller and the headset were measured every 0.1 sec.

3.3 | Reference frame and target location

The reference frame and target positions are shown in Figure 6. The subject holds the controller with both hands, puts it over the left and right greater trochanter respectively, and pulls the trigger button. These controller positions over the right and left greater trochanters are denoted by GT_L and GT_R , the line passing GT_L and GT_R is defined as x axis, the midpoint is defined as origin; the vertical bisector is defined as y axis, and the line that passes the origin normally to xy plane is defined as z axis (In this paper, the right-handed coordinates system is adopted so as to easily distinguish between right and left from the subject's viewpoint; median axis to the pelvis corresponds to 0° on xy plane. That is, the leftward direction relative to the subject's median axis to the pelvis is considered positive). Next, the subject pushes the trigger button at the shoulder joint flexion of 90° and horizontal abduction-adduction of 0° in the sitting position; in this posture, the upper limb length l_a , shoulder height h_s , and eye level h_e are derived from the coordinates of the headset and controller. Using these measured values, 10 candidates for target position for hand position adaptation and head position adaptation were set as shown by the broken circles in Figure 6.

3.4 | Indication of target and body positions

Many USN patients have an impairment of cognitive abilities and judgment. For that reason, as in our previous study,¹⁵ we did not use conventional avatars to represent the head or hand; instead, the hand with the controller or the head with the headset was represented with an opaque sphere (symbolic body, SB), and the target was represented by a translucent sphere (Figures 7 and 8), to make position error easily visible. Besides, to implement an open-loop control in their motor unit movement, the home position was defined as a circle with a radius of 20 cm around the origin (pelvis center), and the target disappears when motor units (controller or headset) go out of the home position.

3.5 | Hand position adaptation

In the hand position adaptation task, conventional PA with prism glasses is implemented in a way similar to that in our previous study¹⁵ or that proposed by Cho et al.¹³ using VR. The difference from those existing methods is that a home position was introduced to implement open-loop control in the subject's upper limb movement.

1. $n = 10$ candidates are generated for the target position (Figure 6).
2. When the subject holds the controller by his dominant hand and moves it to the home position, one of n target position candidates is selected at random, and the target is displayed (Figure 7A).
3. The selected target position is removed from the candidates, and $n = n - 1$ is set.
4. The subject moves, as quickly as possible, the controller in his dominant hand to the position where SB and the target are considered to overlap. When the controller goes out of the home position, the target disappears (Figure 7B).
5. When the subject has moved the controller to the position where SB and the target are considered to overlap and pulled the trigger button, SB and the target are displayed. SB display position is at the controller position in the no shift condition (Figure 7C) or at the position rotated by $\theta = -15^\circ$ around the origin from the SB position of the no shift condition in the shift condition (Figure 7D). The subject confirms the position error between SB and the target.
6. When the trigger button is released, SB and the target disappear (Figure 7E).

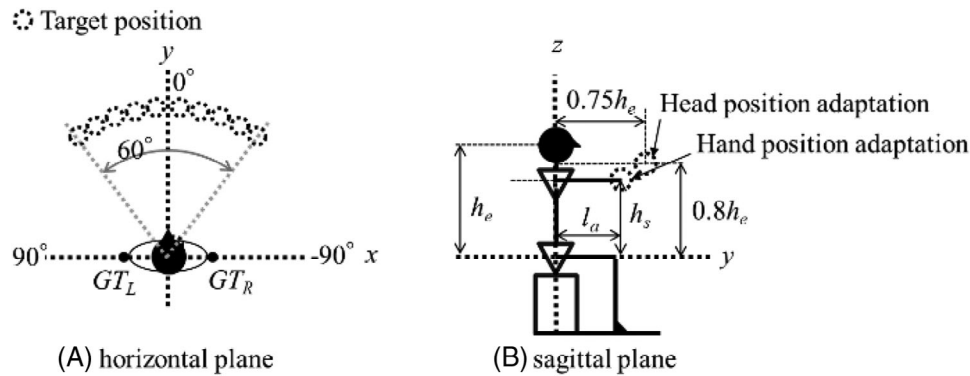


FIGURE 6 Body measurements and target positions.

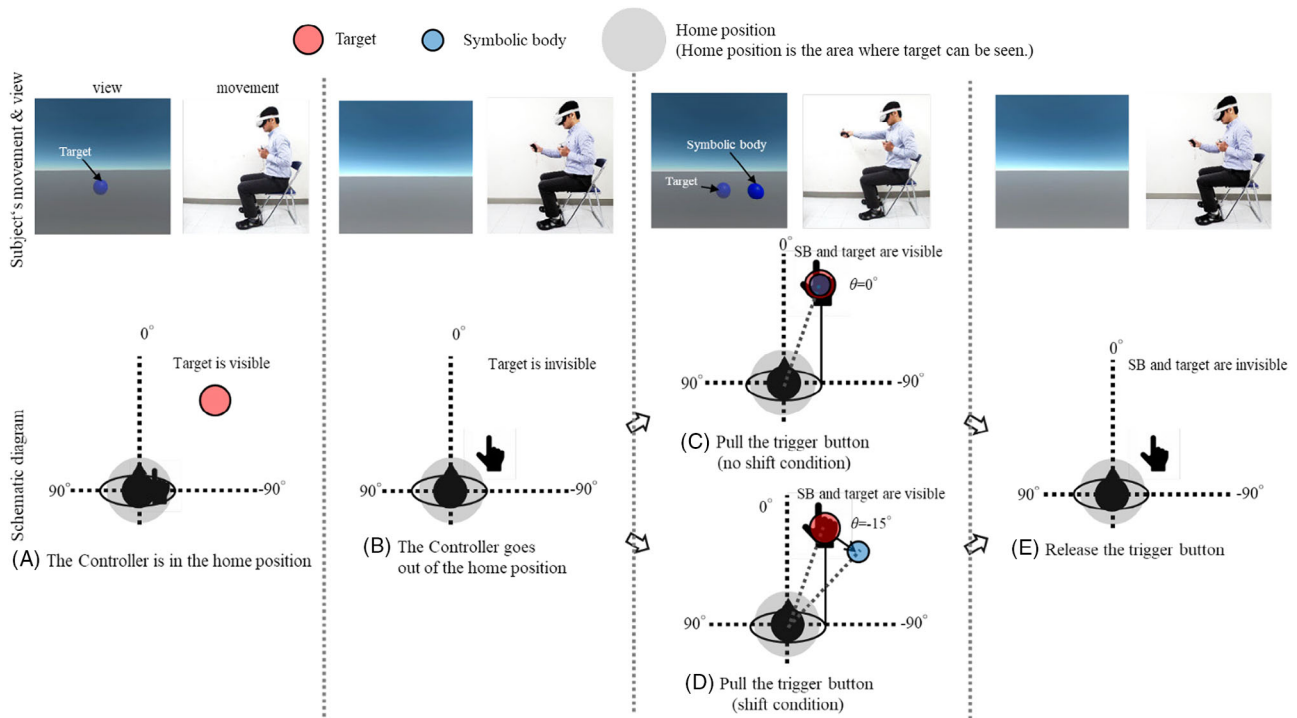


FIGURE 7 Schematic diagram of hand position adaptation trial. [Color figure can be viewed at wileyonlinelibrary.com]

7. If $n > 0$, the procedure returns to (1); if $n = 0$, the trial is terminated.

3.6 | Head position adaptation

In the head position adaptation task, PA is generated for a cluster of motor units that make up the upper trunk system.

1. $n = 10$ candidates are generated for the target position (Figure 6).
2. When the subject moves the headset to the home position, one of n target position candidates is chosen at random, and the target is displayed (Figure 8A).
3. The selected target position is removed from the candidates, and $n = n - 1$ is set.
4. The subject moves, as quickly as possible, the headset to the position where SB and the target are considered to overlap. When the headset goes out of the home position, the target disappears (Figure 8B).
5. When the subject has moved the headset to the position where SB and the target are considered to overlap and pulled the trigger button, SB and the target are displayed. SB display position is 50 cm ahead of the headset in the no shift condition (Figure 8C) or at the position rotated by $\theta = -15^\circ$ around the origin from SB position of the no shift condition in the shift condition (Figure 8D).

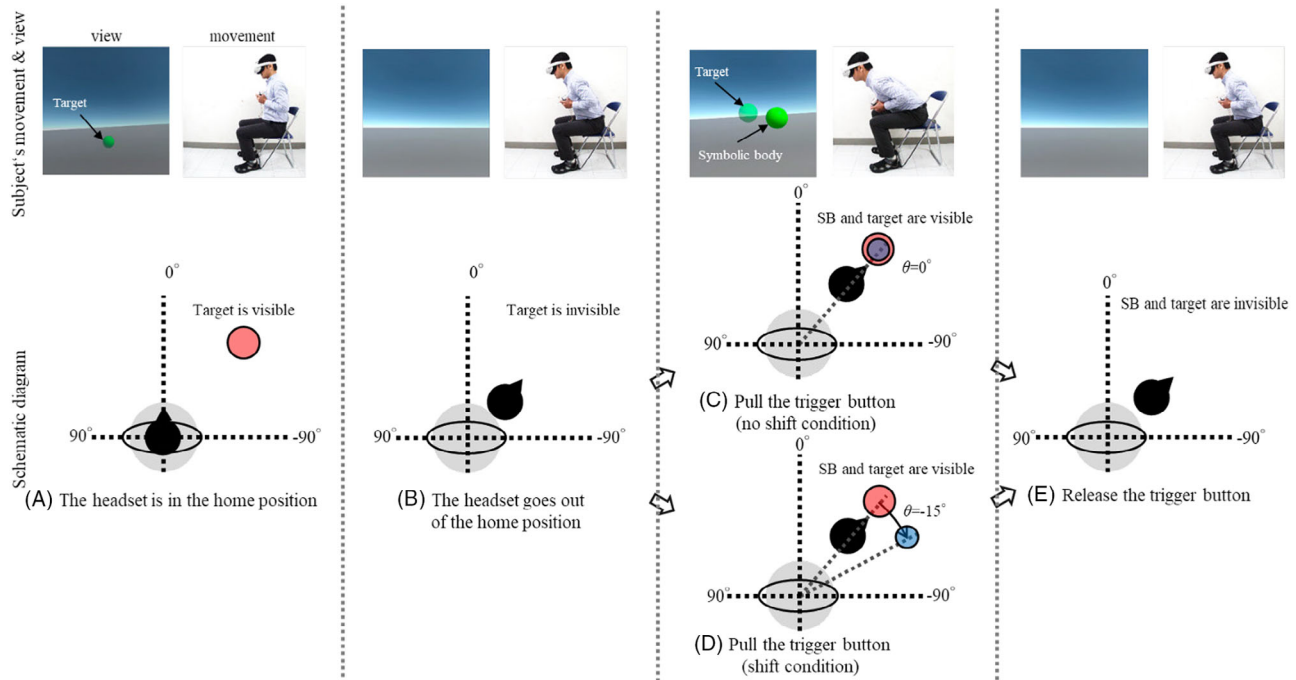


FIGURE 8 Schematic diagram of head position adaptation trial. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/cj.12441)]

6. When the trigger button is released, SB disappears (Figure 8E).
7. If $n > 0$, the procedure returns to (1); if $n = 0$, the set of trials is terminated.

4 | EXPERIMENTS

4.1 | Ethical review

The experiments were approved by the Ethical Review Board of Okinawa College of Rehabilitation and Welfare in accordance with the Declaration of Helsinki. Besides, the subject gave their informed consent.

4.2 | Subjects

The subjects for hand position adaptation were healthy persons without a history of central nervous system disease. The exclusion criteria were an Edinburgh Handedness Inventory of 80 or less and pain in motor units, making the performance of the task difficult. Fifteen subjects were selected, of which 1 was excluded.

The subjects for head position adaptation were healthy persons without a history of central nervous system disease. The exclusion criterion was pain in motor units making the performance of the task difficult. Fifteen subjects were selected, of which none matched the exclusion criterion.

4.3 | Experimental conditions

In both hand and head position adaptation, the experiments included a continuous series of 6 no-shift sets, 6 shift sets, and 6 no-shift sets. The shift angle was set to -15° as recommended by Michel et al.¹⁷

4.4 | Instruction

The following task instructions were given.

4.4.1 | Hand position adaptation

“When you bring your hand to your chest, a translucent blue sphere will appear. This is the target. Quickly move your hand to the target and grasp the controller. An opaque blue sphere will appear at your hand’s position just as long as you are grasping the controller. Repeat your movement so that the two spheres get close to each other as much as possible.”

4.4.2 | Head position adaptation

“Put our hand at your chest. When you move your head just above your pelvis, a translucent green sphere will appear. This is the target. Quickly move your head toward the target and grasp the controller. An opaque green sphere

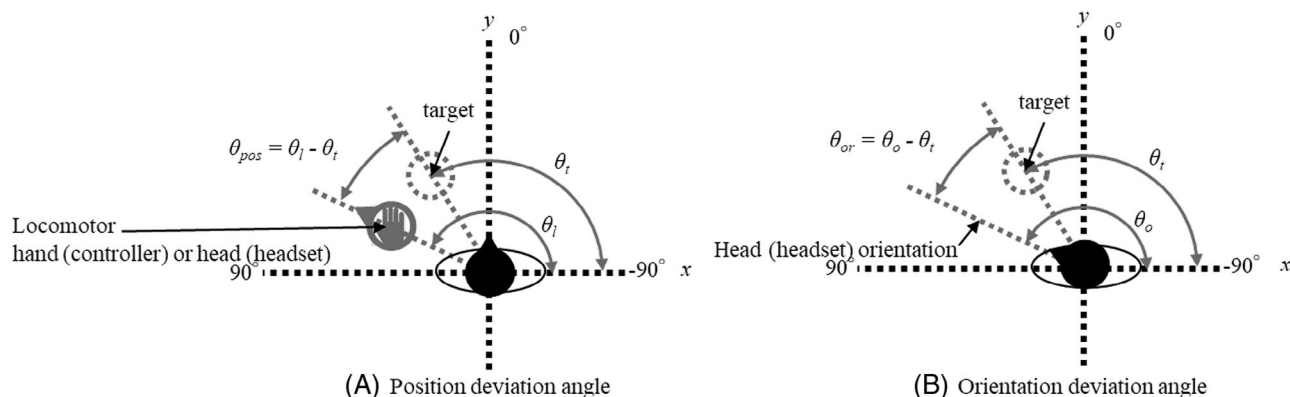


FIGURE 9 Measurement indices.

will appear ahead of your head just as long as you are grasping the controller. Repeat your movement so that the two spheres get close to each other as much as possible.”

4.5 | Measurement indices

The angle θ_{pos} between the target and the motor unit was used as an index of shift adaptability. The angle θ_{or} between the target and the headset was used as an ancillary index to estimate the subject's posture during task performance (Figure 9).

4.6 | Data analysis

The following statistical processing was applied to both hand and head position adaptation tasks.

Average values of θ_{pos} and θ_{or} were derived for each subject in every set, and boxplots were drawn within the interquartile range. Next, the Kruskal–Wallis test was applied to sets 6–18, the last 6 no-shift sets being used as a baseline. As a result, a significant difference at $p < 0.01$ was found only for θ_{pos} in hand position adaptation and for both θ_{pos} , θ_{or} in head position adaptation. Then, the Steel test was applied to the data that showed a significant difference in the Kruskal–Wallis test, and an asterisk(*) was indicated where a significant difference existed (Figure 10). Next, Spearman's rank correlation coefficient was calculated for set-average values in hand and head position adaptation tasks. The coefficient was 0.95 ($p < 0.001$).

In order to explore the items showing correlation with de-adaptation rate, Spearman's rank correlation coefficient was calculated for slopes between sets 12 and 13 showing de-adaptation rate and slopes between sets 6 and 7–12 showing adaptation rate; thus, correlation matrices were obtained (Tables 1 and 2).

The targets were generated according to the body size of each subject (Figure 6). In order to compare the data among different subjects, position coordinate data of the controller and headset were normalized, and scatterplots were generated as explained below. With the hand position adaptation, x , y coordinates of the controller and headset were divided by l_a , and a scatterplot was generated for 6 and 12 sets (Figure 11A). With the head position adaptation, xy coordinates of the controller and headset were divided by h_e , and a scatterplot was generated for sets 6 and 12 (Figure 11B). R ver. 4.03 was used for statistical processing.

5 | DISCUSSION

In this section, we assume a typical left USN patient with postural imbalance toward the backward right (non-neglect side) and discuss rehabilitation to correct that abnormal posture. Conventional PA only applies to the limbs visible through the prism glasses, and researchers have focused on the highly dexterous upper limbs. In our previous study, the prism glasses were replaced with a VR device, and the body was represented by SB in virtual space, which made possible PA for the head orientation.¹⁵ On the other hand, quick movement is impossible in head position adaptation because of the large mass of the upper trunk system. In the case of low motion, the subject can estimate the distance from his head to a target by watching the target. For this reason, the subject conducts a closed-loop control instead of an open-loop control required for PA, as explained in Section 2.2. (Figure 12A). Consequently, to generate PA in slow motor units, such as the upper trunk system, the home position was introduced as a target's visible area, and targets were made invisible for motor units outside of this area. By making a target and motor unit invisible at the beginning of motion, the subject cannot perform closed-loop control and has to resort to open-loop control even if his motion is slow (Figure 12B).

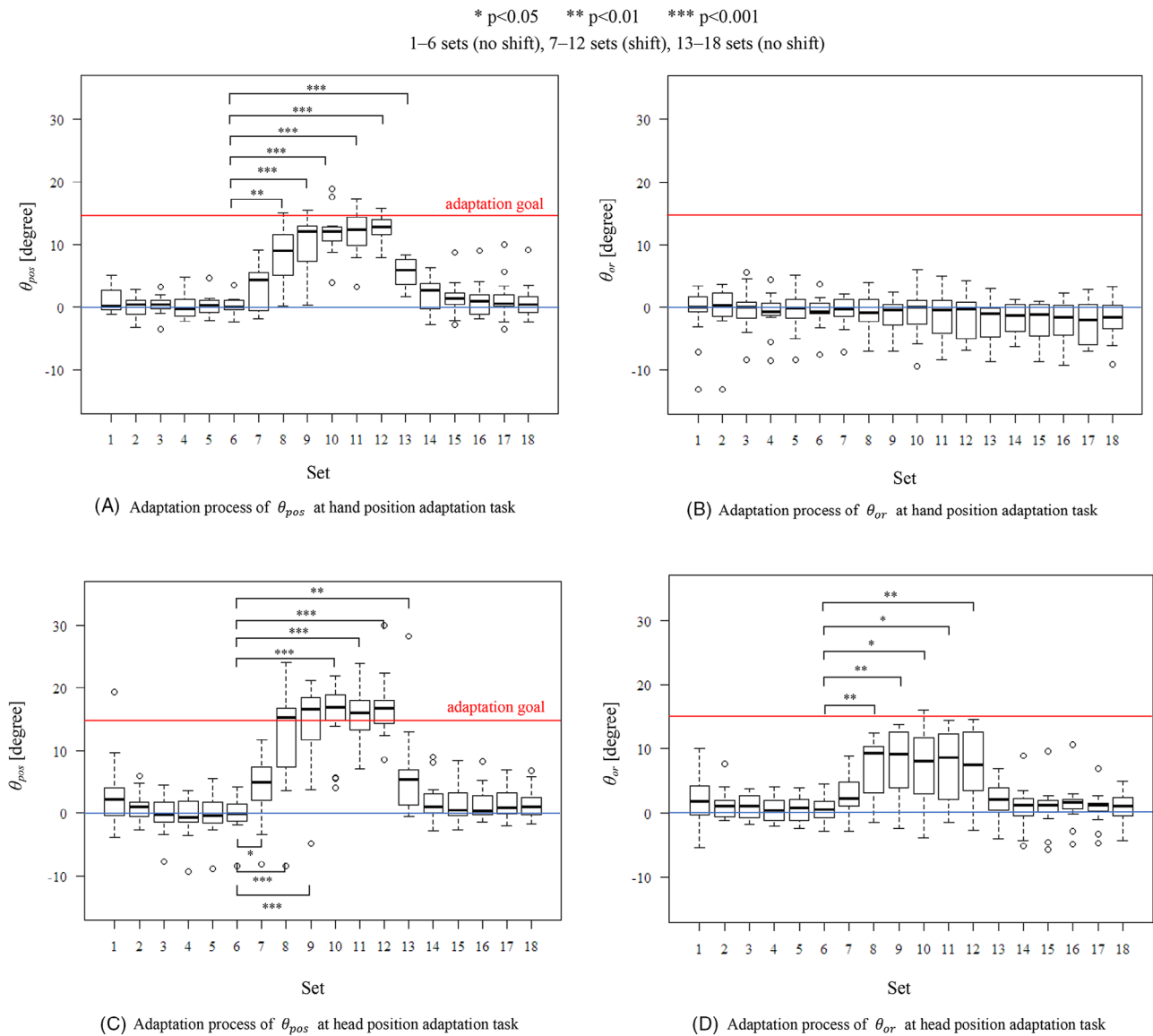


FIGURE 10 Process of hand position adaptation and head position adaptation. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 1 Correlation coefficient matrix of slopes between sets of hand position adaptation.

Variables	Mean	SD	1	2	3	4	5	6	7
1. Slope between sets 6 and 7	−2.99	3.02	–						
2. Slope between sets 6 and 8	−3.85	2.3	0.77**	–					
3. Slope between sets 6 and 9	−3.24	1.47	0.45	0.81***	–				
4. Slope between sets 6 and 10	−2.94	0.91	0.12	0.37	0.63*	–			
5. Slope between sets 6 and 11	−2.35	0.65	0.3	0.49	0.64*	0.75**	–		
6. Slope between sets 6 and 12	−2.09	0.29	0.54*	0.53	0.57*	0.58*	0.79***	–	
7. Slope between sets 12 and 13	7.28	3.33	−0.38	−0.5	−0.48	−0.13	−0.52	−0.63*	–

Firstly, we confirmed that the prism adaptation phenomenon equivalent to that of conventional PA occurred in the head position adaptation task, which is the proposed method. To confirm the aforementioned, we compared the

head position adaptation task with the hand position adaptation task in which the conventional PA was implemented in VR. For this purpose, we discuss the PA task index θ_{pos} . We checked for a significant difference between the last 6

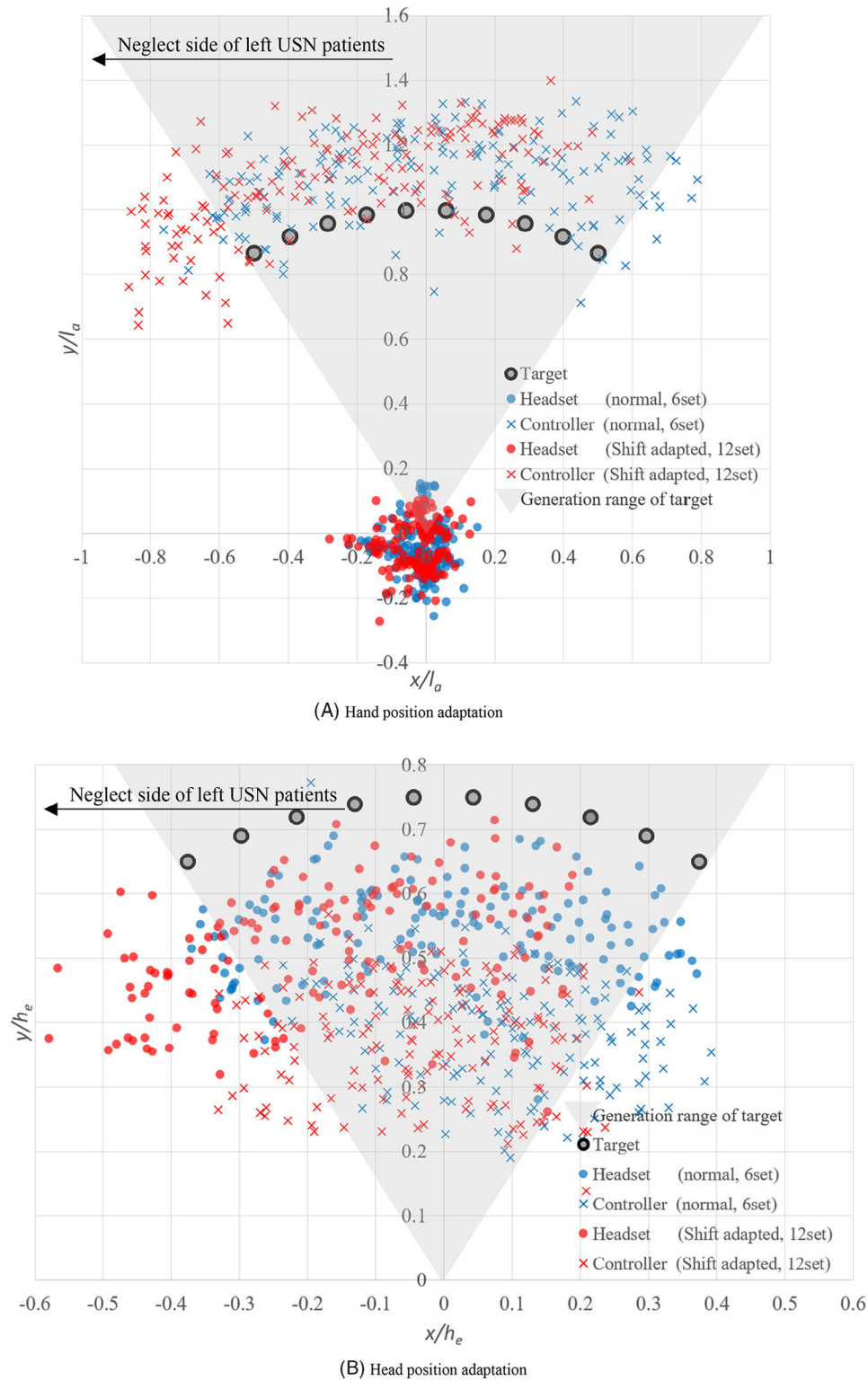
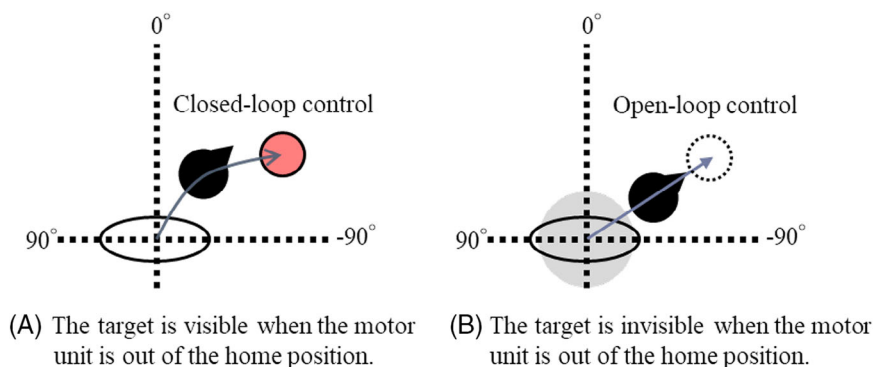


FIGURE 11 Position of controller and headset before and after adaptation in horizontal plane. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/eq.12441)]

TABLE 2 Correlation coefficient matrix of slopes between sets of head position adaptation.

Variables	Mean	SD	1	2	3	4	5	6	7
1. Slope between sets 6 and 7	−4.28	3.76	–						
2. Slope between sets 6 and 8	−6.23	3.72	0.62*	–					
3. Slope between sets 6 and 9	−4.67	1.97	0.48	0.66**	–				
4. Slope between sets 6 and 10	−3.91	1.46	0.15	0.58*	0.74**	–			
5. Slope between sets 6 and 11	−3.22	1.21	−0.13	0.18	0.24	0.45	–		
6. Slope between sets 6 and 12	−2.85	1.15	−0.05	0.27	0.15	0.29	0.9***	–	
7. Slope between sets 12 and 13	10.82	6.79	0.15	0.14	−0.05	−0.11	−0.58*	−0.43	–

**FIGURE 12** Difference in motion control by setting target visible or invisible. [Color figure can be viewed at wileyonlinelibrary.com]

no-shift sets and subsequent sets in both hand and head position adaptation tasks. As a result, a significant difference was found between sets 6 and 8–13 in the hand position adaptation, and between sets 6 and 7–13 in the head position adaptation (Figure 10A,C). In the last 12th shift set, an average of θ_{pos} was 12.69° in hand position adaptation and 16.88° in head position adaptation, thus getting close to the adaptation goal of $+15^\circ$, which is indicative of shift-adapted motion of the subject to cancel out the SB shift angle of -15° . An overshoot trend observed in the head position adaptation can be attributed to the fact that the upper trunk system, as distinct from the upper limbs, is not involved in alignment tasks usually, which makes accurate motion control more difficult. In addition, Spearman's rank correlation coefficient was 0.95 ($p < 0.001$) in both tasks, which is indicative of similar adaptation-deadaptation processes. From these results, we conclude that shift adaptation and de-adaptation similar to conventional PA were implemented in the head position adaptation task.

Furthermore, we consider the index θ_{or} that was not clearly specified as a task. In the hand position adaptation, no significant difference was shown between the 6 sets and all subsequent sets (Figure 10B). On the other hand, in the head position adaptation, the head orientation tended toward the left side (neglect side), where a significant difference was observed (Figure 10D). A big dispersion was observed in sets 8–12 because some subjects turned their heads in the direction of their upper trunk system motion,

while the other subjects looked in the direction of the target display. In the head position adaptation task, head orientation was not specified in the instructions, but more than half of the subjects turned their heads to the left (neglect side), which was a side effect.

In the next, we analyze postures during shift adaptation and de-adaptation (Figure 11). With PA using prism glasses, posture analysis was not carried out because time-course changes in the coordinates of body parts cannot be recorded.^{8–10,12} Similarly, even in previous studies on VPA,^{11,13,14} the subjects' postures were not measured and analyzed. Therefore, we conducted posture analysis based on the scatterplots in which controller and headset positions were projected onto a horizontal plane. Henceforth, the last 6 no-shift sets and 12 shift sets are considered shift-unadapted and shift-adapted, respectively. Firstly, we consider the hand position adaptation task. The controller position shifted to the left (neglect side) of the target in the shift-adapted state, while the headset was located behind the origin in both shift-unadapted and shift-adapted state (Figure 11A). This phenomenon can be explained by the fact that the subjects unconsciously drew their head backward to recognize the whole target generation area extending 30° to the right and left. Drawing the head backward is not desirable in terms of postural correction in USN patients who tilt their heads and bodies backward on the non-neglect side (Figures 1 and 2). In the hand position adaptation task, only one body part—the

upper limb—shifted to the front left. On the other hand, in the head position adaptation task, both the headset and controller shifted forward to the left (neglect side) of the target in a shift-adapted state (Figure 11B). By applying these results to left USN patients (Figures 1 and 2), We conclude that there is the possibility that the USN patient's upper trunk system will be attracted to the front left (neglect side) during the head position adaptation task, therefore alleviating the abnormal posture.

Followingly, we consider the factors affecting the de-adaptation process. In USN rehabilitation, it is desirable that movements toward the neglect side persist after switching from shift to no-shift conditions, that is, the de-adaptation process is slow. As can be seen from the experimental results, data dispersion is relatively high in the 13th set (the first repeated no-shift set) because of the difference among the subjects in the de-adaptation rate (Figure 10A,C). In order to explore the items showing correlation with the de-adaptation rate, correlation analysis was applied to slopes between sets 12 and 13 showing the de-adaptation rate, and between sets 6 and 7–12 showing the adaptation rate (Tables 1 and 2). A significant correlation of -0.63 was found between slopes of sets 12, 13, and 6, 12 in hand position adaptation; also, a significant correlation of -0.58 was found between slopes of sets 12, 13, and 6, 11 in head position adaptation. These results suggest that subjects with rapid shift adaptation tend to show rapid de-adaptation, and vice versa. As regards the causes of such a difference in de-adaptation rate, we assume that the subjects showing rapid shift adaptation discerned condition changes and made adjustments consciously, while the subjects showing slow shift adaptation did not discern condition changes and made adjustments unconsciously. These results suggest that the de-adaptation rate can be decreased not by generating a one-time shift but by gradually increasing the shift amount to make adaptation unconscious. This shift amount cannot be gradually increased when using prism glasses or other optical devices, but this can be done easily with a VR system, which is a great advantage. In addition, with prism glasses, the shift direction is limited to simple rotation. In contrast, VR allows shifting in any direction and at any distance. Using postural analysis in the horizontal plane and SB as proposed in this study, posture can be corrected by setting an appropriate shift direction and shift amount of SB, based on the measurement of the patient's postural imbalance.

Finally, we consider the applied significance of the head position adaptation. Redding et al. reported that when PA is implemented on one upper limb, transfer of the adaptation effect to the other limb is limited,¹⁰ and one can assume that transfer from any body part to another body part is limited as well. Therefore, PA tasks must be assigned

directly to a target body part. Conventional PA-applicable body parts are normally limited to the limbs^{5,8–14,16}, on the other hand, our previous research¹⁵ could be extended to individual motor units capable of moving quickly, such as head orientation. In the present study, body position is represented by SB, and a visible area is provided for targets; this made it possible to extend PA to a cluster of multiple motor units, such as the upper trunk system, that cannot move quickly. In addition, we proposed applying the developed system to postural correction in USN patients. Additionally, applications of the PA proposed in this study are not limited to rehabilitation. Recently, researchers have explored the detection of cerebellar disorders using the fact that patients with such disorders experience difficulties in prism adaptation.¹⁸ Such studies have dealt only with the upper limbs so far, but PA extension to the whole body proposed in this study makes it possible to detect cerebellar disorders concerning body parts strongly affecting balance, such as the upper trunk system. Furthermore, As regards PA with the limbs, fundamental medical and psychological studies have been conducted to clarify adaptation principles or to examine an influence on spatial perception through experiments with healthy subjects.^{10–12} However, there are no studies, except for the present study and our previous study,¹⁵ that consider adaptation principles in the case of PA concerning body parts other than the limbs. PA extension achieved in this study makes possible further clarification of adaptation principles for the whole body.

6 | CONCLUSION

The application range of conventional PA was limited to the visible area through prism glasses, that is, the limbs, especially one upper limb. In this study, a VR device is used to introduce SB for symbolic representation of body position and define a visible area for targets. With the introduction of the concept of SB and target visible area, the developed system can generate PA not to individual body parts but to a collection of multiple motor units such as the upper trunk system, and analyze posture before and after PA. Besides, the experimental results on healthy subjects using a developed system showed that head adaptation can generate PA in the upper trunk system and shift the head front left (the neglect side in left USN patients). The head position adaptation task promises not only the postural correction effect in USN patients but also various applications, such as the detection of cerebellar disorders. In the future, we will use the developed system in experiments with postural correction with appropriate direction and magnitude of shift, based on posture measurement in USN patients.

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